

◆ WEEK 2 — SESSION 4

Compute Layer (EC2 + Instance Types + AWS Lambda)

◆ WHAT IS COMPUTE?

Simple Definition

Compute means the processing power used to run applications and systems.

In AWS, compute allows businesses to:

- Run websites
- Process data
- Host applications
- Handle users
- Execute software

Think of compute as:

“The engine that powers applications.”

◆ WHY CLOUD COMPUTE MATTERS

Before cloud computing:

- Companies bought expensive servers
- Setup took weeks or months
- Scaling was difficult
- Hardware maintenance was costly

With AWS:

- Servers can launch in minutes
 - Resources scale instantly
 - Businesses only pay for what they use
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◆ AMAZON EC2

What is EC2?

Amazon EC2 is a service that provides virtual servers in the cloud.

EC2 stands for:

Elastic Compute Cloud

WHAT EC2 ALLOWS YOU TO DO

With EC2, you can:

- Create virtual machines
 - Install software
 - Run applications
 - Host websites
 - Store and process data
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WHY IT IS CALLED “ELASTIC”

Elastic means:

Resources can increase or decrease when needed.

Example:

- More users visit your website
- AWS can increase compute power
- Traffic drops later
- Resources reduce automatically

This helps save money and improve performance.

◆ EC2 REAL-WORLD EXAMPLES

Businesses use EC2 for:

- Banking applications
- E-commerce websites

- Company portals
 - APIs
 - Enterprise systems
 - Development environments
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◆ EC2 INSTANCE TYPES

What is an Instance Type?

An instance type defines:

- CPU power
- Memory (RAM)
- Storage performance
- Network speed

Different workloads require different resources.

◆ WHY INSTANCE TYPES MATTER

Choosing the correct instance type:

- Improves application performance
 - Reduces costs
 - Prevents overprovisioning
 - Optimizes workloads
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◆ COMMON INSTANCE FAMILIES

1. General Purpose

Balanced CPU and memory.

Used for:

- Web servers
 - Small databases
 - Business applications
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2. Compute Optimized

High processing power.

Used for:

- Gaming servers
 - Analytics
 - Scientific workloads
 - High-performance applications
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3. Memory Optimized

Large RAM capacity.

Used for:

- Big databases
 - Real-time analytics
 - In-memory applications
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4. Storage Optimized

High-speed storage performance.

Used for:

- Data warehousing
 - Logging systems
 - Large storage workloads
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5. GPU Instances

Specialized graphics processing.

Used for:

- Artificial Intelligence
- Machine Learning
- Video rendering
- Graphics workloads

◆ SIMPLE ANALOGY

Different EC2 instances are like different vehicles:

- Small car → Everyday tasks
- Truck → Heavy workloads
- Sports car → High-speed performance
- Bus → Large capacity workloads

AWS provides different “engines” for different business needs.

◆ INTRODUCTION TO AWS LAMBDA

What is Lambda?

AWS Lambda is a serverless compute service.

It allows you to run code:

- Without managing servers
 - Automatically
 - Only when needed
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◆ WHAT “SERVERLESS” MEANS

Serverless does NOT mean there are no servers.

It means:

AWS manages the servers for you.

You focus on:

- Writing code
- Building applications

AWS handles:

- Infrastructure
- Scaling
- Availability

- Maintenance
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◆ HOW LAMBDA WORKS

A trigger happens:

- User uploads image
- API request received
- File added to storage
- Notification event occurs

Lambda automatically:

- Executes code
 - Processes the event
 - Stops when complete
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◆ BENEFITS OF LAMBDA

1. No Server Management

AWS handles infrastructure.

2. Automatic Scaling

Lambda scales automatically.

3. Cost Efficient

You only pay when code runs.

4. Faster Development

Developers focus on applications instead of infrastructure.

◆ EC2 VS LAMBDA

EC2

Lambda

Full server control No server management

Long-running servers Short event-based tasks

Manual configuration Automatic scaling

Greater flexibility Simpler operations

◆ WHEN TO USE EC2

Use EC2 when:

- You need full server access
 - Applications run continuously
 - Custom operating systems are required
 - Workloads need long processing times
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◆ WHEN TO USE LAMBDA

Use Lambda when:

- Tasks are event-driven
 - Workloads are short
 - Automation is needed
 - You want serverless architecture
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◆ KEY TAKEAWAYS

- ✓ Compute powers applications in AWS
 - ✓ EC2 provides virtual servers
 - ✓ Instance types affect performance and cost
 - ✓ Lambda enables serverless computing
 - ✓ AWS allows scalable and flexible infrastructure
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◆ QUICK RECALL QUESTIONS

1. What does compute mean in cloud computing?
2. What does EC2 stand for?
3. Why are instance types important?
4. What does serverless mean?
5. What is the difference between EC2 and Lambda?

◆ FINAL THOUGHT

“Modern cloud architecture is about using the right compute model for the right workload.”